

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 7: Dynamic Choice

Notes on the Chapter

If every author is entitled to one bee-in-the-bonnet chapter, Chapter 7 is mine. In the chapter itself, I describe and then rail against the usual way in which economists deal with choice in dynamic contexts, namely that dynamic choice is simply static choice (and then flawless implementation) of an optimal dynamic strategy. There has recently been enough activity under the rubric behavioral economics so that my rant is no longer so much of a lone cry in the wilderness. But I anticipate that many instructors will regard this as an unnecessary and ill-advised detour from the main line taken in this book.

Obviously, I disagree with this. In the end, I will only use the standard approach to dynamic choice for the remainder of this volume (to the extent that dynamic choice enters the story at all, which is not to a very great extent except for Chapter 16). Alternatives to the standard model will reappear only in Volume III (with the usual "when and if ever" caveat taken as read). But given the importance in economics of contexts in which choices are dynamic or (at least) are made within a larger dynamic-choice setting, I believe it is important in getting the foundations straight to acknowledge that the standard approach comes with some pretty heroic assumptions and that relaxing those assumptions is far from impossible.

My hope, then, is that this chapter, which is pretty "easy reading" compared to most, will make those points.

But if the chapter is easy reading, it is accompanied by Appendix 6, which is anything but. In Appendix 6, I provide a primer on the methods of dynamic programming or, at least, those portions of dynamic programming where the overall reward over an infinite horizon depends asymptotically on what happens in finitely many periods. (For aficionados, this means that I cover the classic results from positive and negative dynamic programming.) The appendix is over 50 pages long and, as I anticipate dynamic programming will be new to most students, it is likely to be quite a slog.

Dynamic programming is, I gather, a topic taken up more often in macro-theory courses than in micro. But it fits in well with some of the themes of Chapter 7 (in particular, the assumption that decision makers have the analytical capabilities required

to find an optimal dynamic strategy). And, anticipating that, in Volume II, infinite-horizon repeated games will be on the agenda, it needs to be covered. So I take it on here. For this reason, "Chapter 7" for me takes a full week of two 105-minute class sessions; one session devoted entirely to the Appendix and some of the problems given there; then a second session on Chapter 7 itself.

A final point is that my approach to the theory of dynamic programming is not exactly standard. The usual formulation with a discounted sum of rewards is not introduced until basic results are given about Bellman's equation and the (sometime) optimality of conserving and unimprovable strategies; from a mathematical perspective, I think the formulation with discounted sum of rewards hides rather than reveals the mathematical structure of the basic results. And a bit of sophistication about probability theory (the law of iterated conditional expectations; at least a head feint in the direction of the Kolmogorov Extension Theorem) is required.

The Appendix provides a number of unsolved problems, each of which has solution provided in the *Student's Guide*. The chapter itself has four problems; the solutions of the first three (all about Strotzian changing tastes) are in the *Guide*; for the fourth (the basic "representation of preference for flexibility"), I direct the reader to my old 1979 paper on the topic. Given the recent interest in the literature concerning hyperbolic discounting, the one problem to assign if you are assigning only one is Problem 7.2.